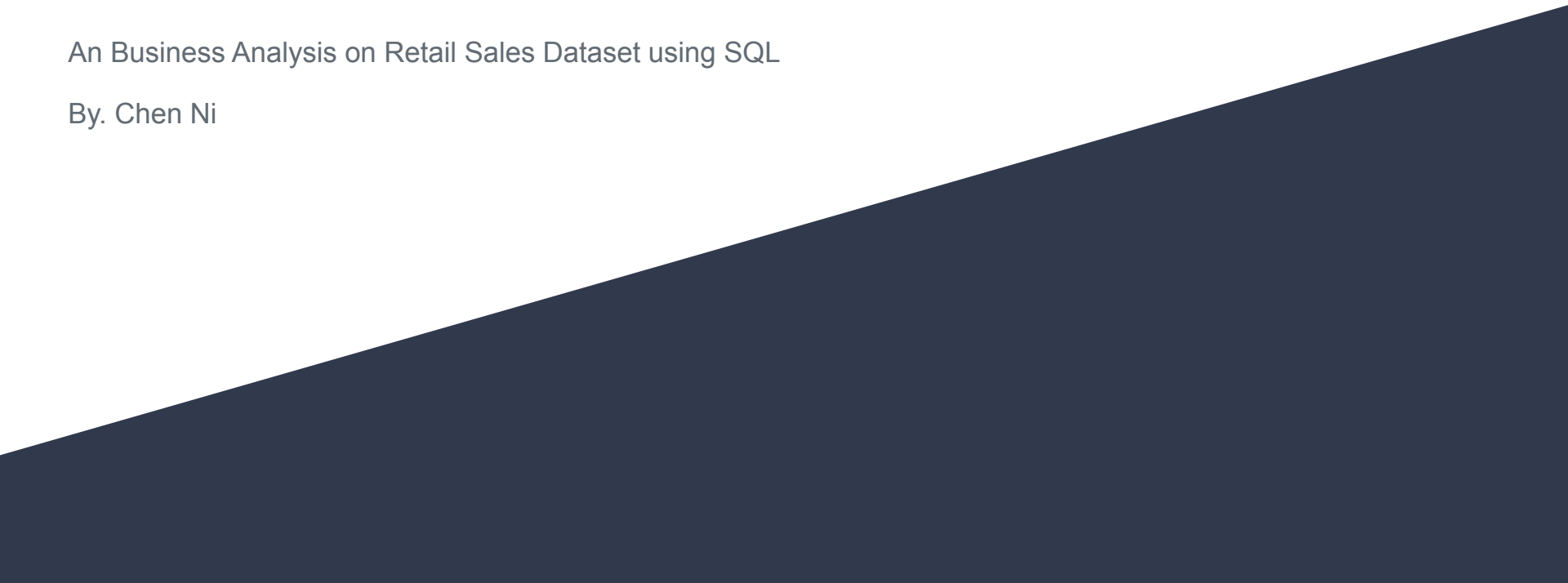


Retail Sales Analysis

An Business Analysis on Retail Sales Dataset using SQL

By. Chen Ni

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Topics

- Background
- Analysis
- Result - Interpretation
- Recommendation

Background

Background

- The objective of The Retail Company is to analyze customer behavior and identify opportunities for growth.
 - Most and least profitable categories
 - Gender breakdown on categories
 - Seasonal pattern in customer purchases
- The dataset provided by the company includes sales data from 2022 through 2023.
- This project initiated The Retail Company's shift to data-driven decisions.
- The retail company mandates an explanation of the context underlying the recommendation.

Analysis

Analysis:

Most and least sale

- Electronic is the category with **most sale**.
- Beauty is the category with **least sale**.

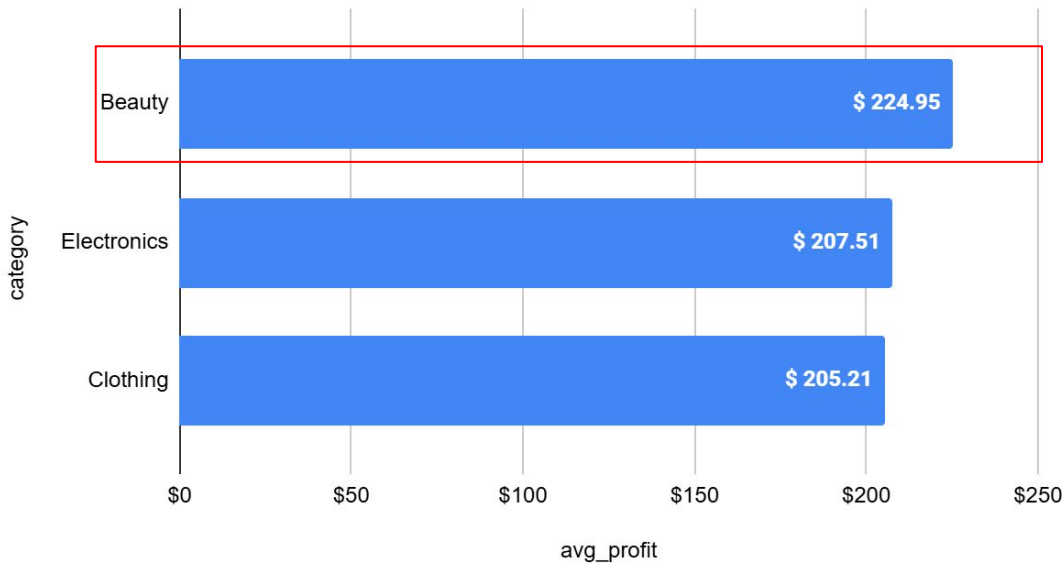


Analysis:

Average profit

- Beauty is the category with **highest average profit** out of the three categories that The Retail Company offered.

Average profit of Category



Analysis:

Gender breakdown

- The Beauty category shows the greatest difference between gender customer groups.

Gender breakdown by Category

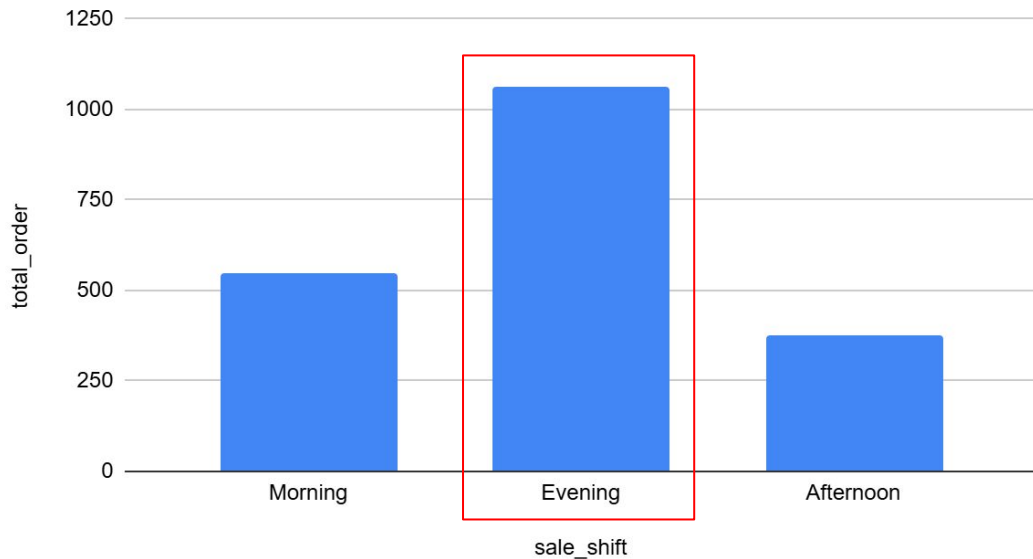


Analysis:

Seasonal pattern

- Most customer place order during evening.

Total order by Sale shift

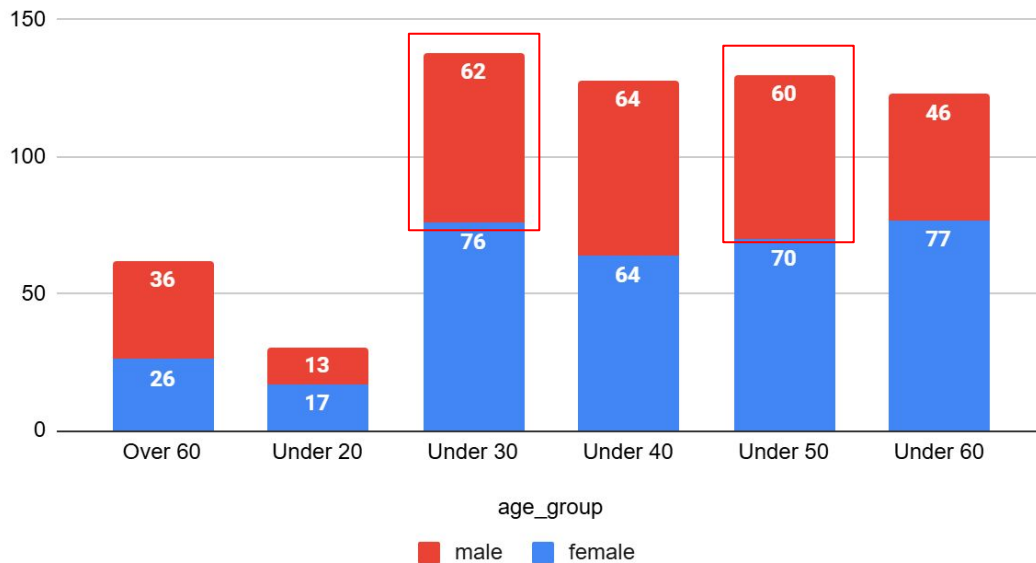


Analysis:

Target customer segment

- Our three largest customer segments are individuals aged 20 to 49.
- Male customers aged 20-29 and 40-49 appear to be a promising market segment to target.

Purchase volume by age group in Beauty category

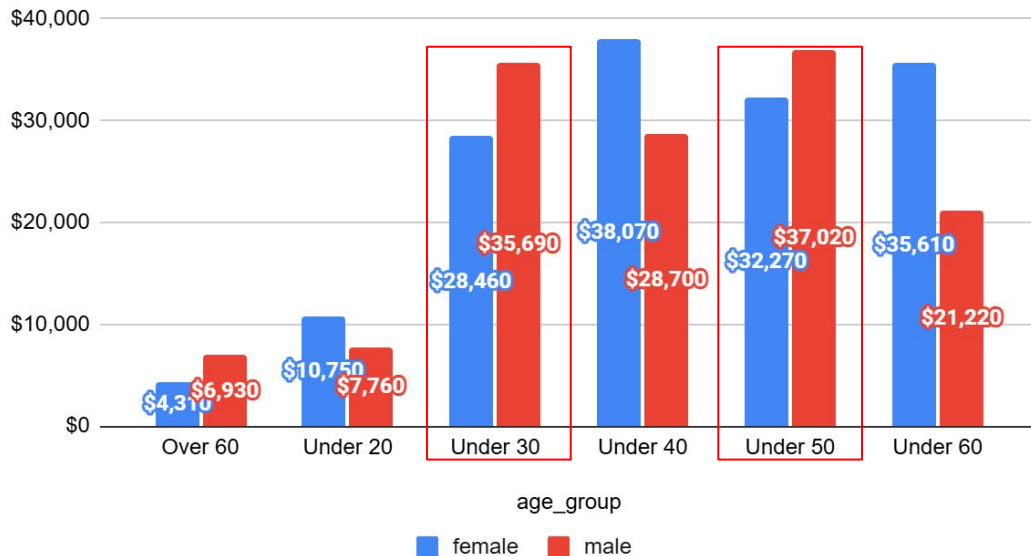


Analysis:

Customer segment spending

- Male customers aged 20-29 and 40-49 are the highest spenders in their respective age groups, outspending their female counterparts.

Total amount by age group in Beauty category



Analysis:

Purchase habit in beauty category

- Both genders make the most purchase decisions during evening hours.

Purchase habit by gender between age 20-49



Result

Result – Analytical Result

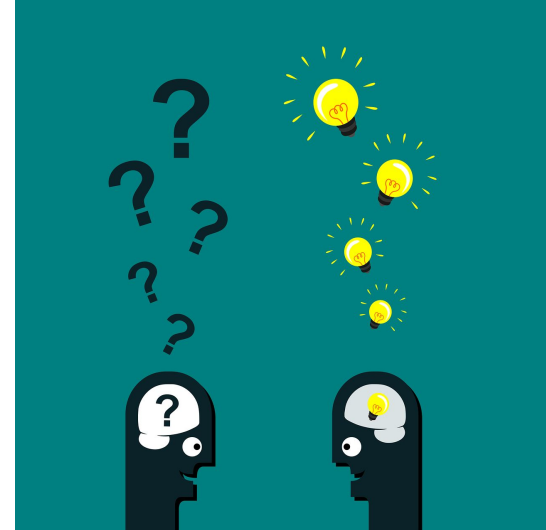
- **What we found:**

- **Beauty Category** bring in the least sale volume but have the highest avg. profit.
- The beauty category is the only category with an **uneven gender breakdown**.
- Most of our customers make their purchase decision during **evening**.
- The **20-49 age group** drives the **highest sales volume** in the beauty category.
- **Male customers aged 20-29 and 40-49** purchase less volume than their female counterpart part but bring in higher sell amount.



Result – Interpretation

- **What should we consider doing?**
 - **Target growth within Beauty Category:** Highest avg. profit compared to other categories
 - **Focus on the male customers in the age segment of 20-49:** In this traditionally female-dominated category, they spend at or above the level of their female counterparts.
 - **Target promotion on customers during evening hours:** Highest sale volume happened during the evening.
 - Focus especially on **Male customers in the age segment of 20-49.**



Result – Interpretation

- **What factors contribute to the increasing consumption of beauty products by men?**
 - “One of the primary drivers behind the men’s skincare boom is the **shift in cultural attitudes towards masculinity and self-care**.”¹
 - “... is important to **maintain an attractive body**, is to feel good about himself that, and is the main motivation, **improve the image seen by the other**, which implies the need to take care of themselves.”²
- **How big is the male beauty market size?**
 - “... global male grooming market is expected to be worth **\$115 billion** by 2028, up from nearly **\$80 billion** in 2022.”³
 - “Estimates vary, but in 2022 the global market was valued at **202.6 billion dollars** and is expected to grow at an annual rate of **8 percent** until 2030.”⁴



¹Kachi, E. (2024, August 30). Understanding the men's skincare boom. BeautyMatter. <https://beautymatter.com/articles/understanding-the-mens-skincare-boom>

²Borgna, I. (2018, March 12). Cosmetics and male consumers. Kosmetika World. <https://www.kosmeticaworld.com/2018/04/04/cosmetics-male-consumers/>

³Broeke, T. van den. (2023, July 19). The male grooming industry is booming. here's why. CNN. <https://www.cnn.com/style/male-grooming-industry-booming/index.html>

⁴Adegeest, D.-A. (2023, August 24). The men's beauty and personal care market is booming. FashionUnited. <https://fashionunited.uk/news/fashion/the-men-s-beauty-and-personal-care-market-is-booming/2023041268947>



Brands globally are targeting male customers in the beauty segment.

Recommendation

Recommendation

- The Retail Company should target male customers aged 20-49 to promote the Beauty category to boost sales.
- Maximize conversions by prioritizing evening promotion durations.